INCH-POUND

MIL-L-49147C(CR)
27 May 1992
SUPERSEDING
MIL-L-49147B(ER)
8 November 1989

MILITARY SPECIFICATION

LENS ASSEMBLY, EYEPIECE
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Lens Assembly Eyepiece, referred to herein as eyepiece, which is part of Viewer, Driver's Night Vision AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.c).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-O-13830	-	Optical Components for Fire Controls Instruments General Specification Governing the Manufacture, Assembly, and Inspection of

STANDARDS

MILITARY

MIL-STD-150	-	Photographic Lenses
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.c).

DRAWINGS

USA COMMUNICATIONS - ELECTRONICS COMMAND

SM-D-771327	-	Lens Assembly Eyepiece
SM-C-771240	-	Magnifier, Tapered Fiber Optic, Viewer

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The eyepiece is mounted to and is part of the Viewer, Driver's Night Vision AN/VVS-2(V).

3.2 Construction. The eyepiece shall be constructed in accordance with SM-D-771327 and as specified herein.

3.3 Qualification. Unless otherwise specified by the contractor or purchase order, Eyepiece Assemblies furnished under this specification shall be products approved as qualified products (see 3.11.3 or 6.3).

3.3.1 Initial production testing. The supplier shall furnish and test eyepieces as specified in the contract or purchase order (see 6.2.f).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Performance characteristics.

3.5.1 Diopter setting. The eyepiece shall be set at -2.0, +0.3/-0.1 diopter as viewed from a distance of 50 millimeters (mm) on axis when installed on the AN/VVS-2(V).

3.5.2 Resolution. The eyepiece shall have a minimum resolution of 20 lp/mm.

3.5.3 Magnification. The on-axis magnification shall be between 3.75 and 4.0.

3.5.4 Transmission. The transmission of the eyepiece shall be 80 percent minimum for the region of spectral output of a P-20 phosphor.

3.5.5 Field of view. The field of view of the eyepiece shall be 42.5 degrees minimum.

3.5.6 Field flatness. The field flatness shall deviate no more than 1/4 diopter over a 46 mm format with a 100 mm radius of curvature for the fiber optic magnifier output.

3.6 Environmental.

3.6.1 High temperature. The eyepiece shall withstand a storage temperature of +68°C without damage or degradation of performance (see 3.11.1 and 3.11.5).

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3.6.2 Low temperature. The eyepiece shall withstand a storage temperature of -51°C without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.3 Temperature shock. The eyepiece shall withstand cycles of sudden changes in temperature between -51°C and +68°C without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.4 Sealing. The eyepiece shall be filled with a mixture of dry nitrogen with 1 mole-percent helium and shall be sealed. The helium leak rate of the sealed eyepiece shall be not greater than $9.9 \times 10^{-4} cm^3/sec$. No damage or degradation of performance is allowed (see 3.11.1 and 3.11.5).

3.6.5 Vibration. The eyepiece shall withstand vibration equivalent to that encountered in tracked vehicles, during 5000 miles of operation without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.6 Shock.

3.6.6.1 Basic shock. The eyepiece shall withstand basic shock of 40g's (see 3.11.6) at 18 milliseconds without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.6.6.2 Weapon shock. The eyepiece shall withstand simulated weapon firing shock of 100 g's at 1.5 milliseconds one half sine pulse without damage or degradation of performance (see 3.11.1 and 3.11.5).

3.7 Treatment and paintings. Unless otherwise specified, the metallic parts of the eyepiece shall be given a treatment and painting per MIL-STD-171.

3.8 Marking. The eyepiece shall be marked in accordance with SM-D-771327.

3.9 Cleanliness. The eyepiece shall meet the cleanliness requirement for optical systems as specified in MIL-O-13830.

3.10 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any referenced specifications or drawings when inspected in accordance with Section 4. Parts and assemblies shall comply with the requirements of MIL-O-13830.

3.11 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.11.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.11.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

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3.11.3 Qualified products. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characterisitcs of the product affected by the changes.

3.11.4 Environmental test. For the purpose of this specification, environmental tests are those discussed in 4.6.

3.11.5 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

3.11.6 "g". "g" is a unitless measure of the quantity of units of force associated with the earth gravity that a object is subject to.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial production tests (4.3)
b. Quality conformance inspection (4.4)
c. Inspection for packaging (4.7)

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4.3 Initial production testing.

4.3.1 Inspection. Each IPT eyepiece shall be inspected in accordance with Table I, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that eyepiece and may be cause for rejection of IPT.

4.3.1.1 Disposition of IPT samples. Disposition of IPT samples shall be as specified in the contract or purchase order (see 6.2.f).

TABLE I. Inspection.

Defects	Requirement paragraph
Parts and materials not specified	3.4
Treatment and painting	3.7
Identification or special marking missing or illegible	3.8
Cleanliness	3.9
Workmanship	3.10

4.3.2 Test. Unless otherwise specified (see 6.2.f) ten eyepieces manufactured with production tooling shall be selected from the first production lot and provided for IPT. Following successful completion of the inspections specified in 4.3.1, each IPT eyepiece shall be subjected to all the tests of Table II. Tests may be performed in any order. Failure of any tests shall be cause for rejection of the IPT.

TABLE II. Initial production testing.

Inspection	Requirement paragraph	Test paragraph
Diopter setting	3.5.1	4.5.1
Resolution	3.5.2	4.5.2
Magnification	3.5.3	4.5.3
Transmission	3.5.4	4.5.4
Field of view	3.5.5	4.5.5
Field flatness	3.5.6	4.5.6
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Sealing	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Shock	3.6.6	4.6.6

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4.4 Quality conformance inspection.

4.4.1 Examination. Each eyepiece shall be inspected in accordance with Table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Group A inspection. Eyepieces that have passed the examination specified in 4.4.1, shall be tested in accordance with Table III in the order listed. Failure of any test shall be cause for rejection of that unit. Eyepieces which pass Group A inspection shall be formed into lots by the contractor.

TABLE III. Group A inspection.

Test	Requirement paragraph	Test paragraph
Diopter setting	3.5.1	4.5.1
Resolution	3.5.2	4.5.2

4.4.2.2 Group B. Group B tests shall be conducted on eyepieces selected from lots which have passed the examination and the Group A test. Sampling shall be in accordance with the contract or purchase order. Tests shall be conducted in accordance with Table IV.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Magnification	3.5.3	4.5.3
Transmission	3.5.4	4.5.4
Field of view	3.5.5	4.5.5
Field of flatness	3.5.6	4.5.6

4.4.2.3 Group C. This inspection shall consist of the tests specified in Table V and shall be performed on lots that have been subjected to and have met Group A inspection. Sampling shall be in accordance with the contract or purchase order.

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TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.5	4.6.5
Shock	3.6.6	4.6.6
Sealing	3.6.4	4.6.4

4.4.2.3.1 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order, (see 6.2.b).

4.4.2.3.2 Disposition of Group C samples. Unless otherwise specified, samples which have been subjected to Group C tests shall be refurbished, pass the requirements of Group A tests, and shipped against the requirements of the contract.

4.5 Operational Test methods. Unless otherwise specified all tests shall be performed at ambient temperature (see 3.11.2).

4.5.1 Diopter setting. Using the test set up in 4.5.2 observe the test target using a diopterscope. Measure the diopter focus with the diopterscope 50 mm from the vertex of the eyelens. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Resolution. A thin fiber optic face plate, with a maximum of 13 micrometer fibers, with a curvature per SM-C-771240, is placed at the normal distance from the eyepiece mounting flange. A high contrast (per MIL-STD-150) USAF 1951 Resolving Power Test Target (clear lines on black background) is placed in contact with the flat side of the fiber optic face plate (see Figure 1). View binocularly through the eyepiece from a 50 mm eye relief. Failure to meet the requirement of 3.5.2 shall constitute failure of this test.

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4.5.3 Magnification. Set up test equipment as shown in Figure 1. Back illuminate the test scale with lambertian P-20 light. Place the diopter scope on the rotary table such that its optical axis is coincident with optical axis of the eyepiece. The rotary table is set with its axis 50 mm from the vertex of the eyepiece. Record the angles read on the rotary table at the ± 3mm positions (H and J).

Average the two angles for ± 3 mm positions and calculate magnification M.

where:

$$M = \frac{254 \text{ mm} \tan \Theta}{3 \text{ mm}}$$

Θ = average of angle for 3 mm field points from axis. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Transmission. Using test set up in 4.5.2 illuminate the fiber optic face plate with P-20 light (diffuse). Measure with a photopic detector. The transmission is the ratio of the reading when light passes through the biocular eyepiece to the brightness of the fiber optic face plate alone. Failure to meet the transmission requirement of 3.5.4 shall constitute failure of this test.

4.5.5 Field of view. Field of view of the eyepiece shall be measured using the test set up in 4.5.2. Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Field flatness. Field flatness of the eyepiece shall be measured using the test set up in 4.5.2 and a diopterscope. Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

4.6 Environmental tests methods. The eyepiece shall be retested to meet the requirements of 3.5.1, 3.5.2, and 3.5.4 after the completion of each environmental test (see 3.11.4). Failure to meet these requirements shall constitute failure as defined in degradation of performance. Temperature chambers shall maintain specified temperature within ± 2°C.

4.6.1 High temperature. The eyepiece shall be subjected to the test of Method 501.1, Procedure I of MIL-STD-810, except that the temperature of Steps 2 and 3 (highest storage temperature) shall be +68°C and Steps 4 and 5 shall be deleted. Step 6 stabilization period shall be a minimum of two hours. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.4, and 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The eyepiece shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -51°C. Steps 2 and 6 stabilization period shall be a minimum of two hours, and Steps 4 and 5 shall be deleted. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.4, and 3.6.2 shall constitute failure of this test.

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4.6.3 Temperature shock. The eyepiece shall be subjected to the test of Method 503.1, Procedure I of MIL-STD-810, except the high temperature chamber shall be maintained at +68°C, the low temperature chamber shall be maintained at -51°C, and Steps 1, 3, and 5 stabilization period shall be a minimum of two hours. Step 1 and 4 shall be at +68°C, step 2 shall be at -51°C. Immediately after each cold cycle, the eyepiece shall be examined for internal condensation. Evidence of moisture or condensation on any internal surface or failure to meet the requirements of 3.5.1, 3.5.2, 3.5.4, and 3.6.3 shall constitute failure of this test.

4.6.4 Sealing. The eyepiece shall be mounted onto a sealed container which simulates the main viewer housing assembly interface and subjected to Method 512.1, Procedure IV of MIL-STD-810 using water as the liquid. The eyepiece shall then be tested for helium leakage with a Varian Model 925-40 Porta-Test or equivalent. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.4, and 3.6.4 shall constitute failure of this test.

4.6.5 Vibration. The eyepiece shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6, Tracked Vehicles, limited to one sweep in the vertical axis only. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.4, and 3.6.5 shall constitute failure of this test.

4.6.6 Shock. The eyepiece shall be subjected to the test of MIL-STD-810, Method 516.2.

a. 40g's at 18 milliseconds, Figure 516.2-1, shall be the basic design test shock, Procedure I.

b. Weapon firings, 100g's using Procedure IV except at 1.5 milliseconds (ms), Figure 516.2-2.

Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.4, and 3.6.6 shall constitute failure of this test.

4.7 Inspection for packaging. Packaging shall be inspected to determine compliance to requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.g).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The eyepiece covered by this specification is intended for use in the Viewer, Driver's Night Vision AN/AVS-2(V), an optical viewing device capable of directing the line of sight to the driver's eyes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Necessary actions by the contractor in the event of a Group C failure (see 4.4.2.3.1).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
d. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105D.
e. MIL-STD-810C is used for environmental tests of section 4.
f. Quantity and schedule of IPT and disposition of IPT samples.
g. Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.11.

6.5 Subject term keyword listing.
Eyepiece
Viewer Driver's

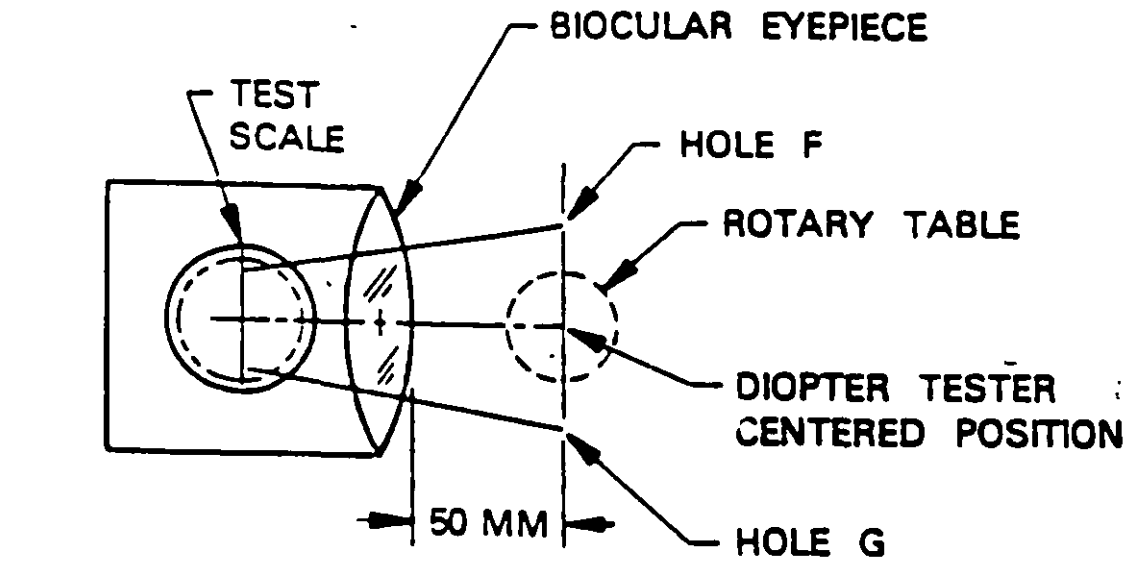
6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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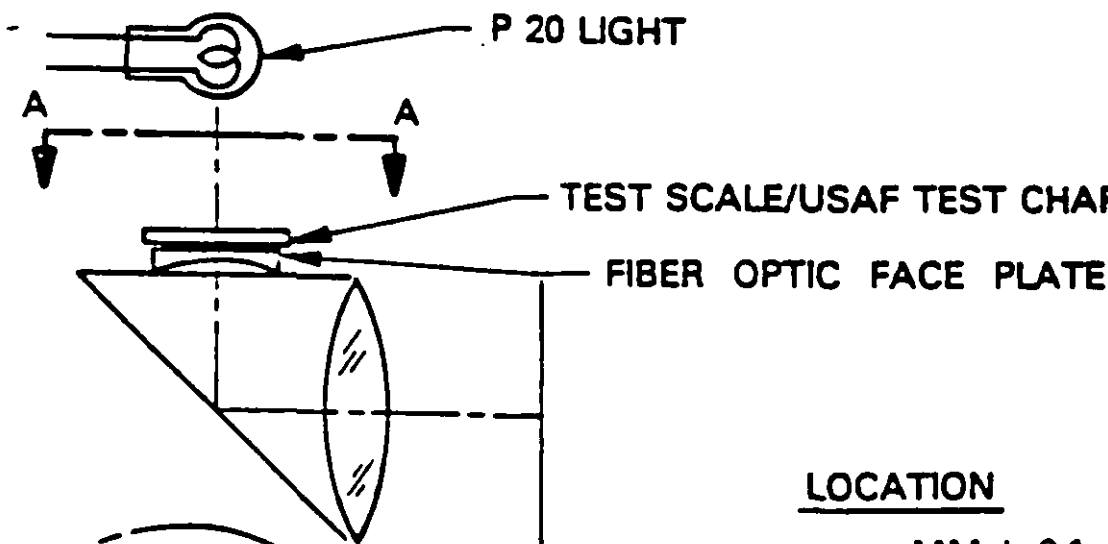
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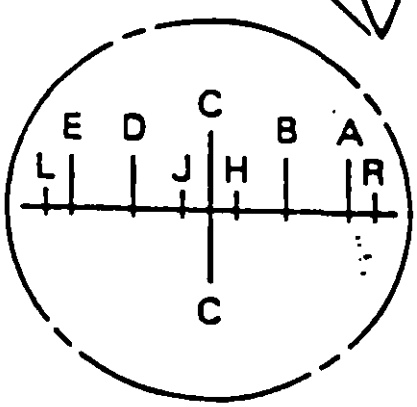
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VIEW A-A





TEST SCALE

LOCATION	MM ± 0.1
R	23.9
A	23
B	10
H	3
C	0
J	3
D	10
E	23
L	23.9

FIGURE 1. EYEPIECE TESTS DIAGRAM

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-49147C(CR)	27 May 1992

3. DOCUMENT TITLE
LENS ASSEMBLY, EYEPIECE VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908) 532-5851	992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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